SOURAV LAI

Backend Developer | PHP | Laravel | CodeIgniter | MySQL | AI Integration
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PROFESSIONAL SUMMARY

Backend PHP Developer with 4+ years of experience at Shreesols designing scalable REST APIs and optimizing large-scale ERP systems covering payroll, HR and support ticketing. Reduced a critical report page's load time from 8 hours to 10-15 seconds using MySQL stored procedures, functions and indexing. Strong in MySQL stored procedures, functions, triggers and query tuning, and in AI engineering: LLM integration (OpenAI, Claude, Gemini), RAG, tool-calling agents and MCP servers, including a RAG support agent on Cloudflare Workers AI and an AI-moderated dating platform.

TECHNICAL SKILLS

Programming: PHP 8, JavaScript, jQuery, SQL, HTML5, CSS3
Frameworks: Laravel, Livewire, CodeIgniter 4, Core PHP MVC, Bootstrap
Databases: MySQL (Advanced): Stored Procedures, Functions, Triggers, Indexing, Query Optimization; MariaDB, Redis
API Development: RESTful APIs, Webhooks, WebSockets (Ratchet, ReactPHP), Authentication, RBAC, OAuth
AI: LLM Integration (OpenAI, Claude, Gemini), RAG, Vector Embeddings, Tool Calling, AI Agents, MCP Servers, Prompt Versioning
Cloud: Cloudflare Workers, Workers AI, Vectorize, D1, KV, R2, Turnstile, AWS (S3, SES)
Integrations: Telegram Bot API, WhatsApp API, Agora, LiveKit, Razorpay, Stripe, PayPal, Paytm, Firebase Cloud Messaging
DevOps: Docker, Git, GitHub, CI/CD, Apache, PHP-FPM
Practices: PHPUnit, Performance Optimization, Security Best Practices, Code Reviews, SDLC
Spoken Languages: English, Hindi, Bengali

PROFESSIONAL EXPERIENCE

Software Developer | Shreesols, Kolkata — Jul 2022 – Present

- Reduced the execution time of a critical report page from 8 hours to 10-15 seconds by implementing optimized MySQL stored procedures, functions and indexing strategies.
- Develop and maintain ERP modules including payroll management, HR and support ticketing using PHP, Laravel and CodeIgniter, with a focus on performance, scalability and security.
- Built RESTful APIs and web services connecting application components with external systems and a Flutter mobile app.
- Designed and optimized MySQL databases with stored procedures, functions and triggers (conditional logic and loops), indexing, normalization and query optimization.
- Implemented authentication and authorization, including user authentication, role-based access control (RBAC) and OAuth integration.
- Collaborated with front-end developers on integration and maintained code quality through coding standards and code reviews across the full SDLC, from requirements to deployment.

KEY PROJECTS AT SHREESOLS

FUSE ERP | PHP, MySQL

- Developed a comprehensive ERP system with HR, Payroll, Inventory Management and CRM modules to streamline business processes.

Fuse Mobile App (Backend) | PHP, CodeIgniter

- Built RESTful APIs for a Flutter mobile app covering data retrieval, user authentication and database interaction.

Signer | PHP, Laravel, JavaScript

- Created a digital signature platform with user authentication, document management and encryption for secure document signing.

Moodle LMS Customization | PHP, MySQL, Moodle

- Customized and extended Moodle with course management, user enrollment, grading and reporting features.

PROJECTS

LoveBridge – AI-Moderated Dating Platform | CodeIgniter 4, Laravel, NativePHP, MySQL, Telegram Bot API

- Built a dating platform delivered as a Telegram bot, Telegram Mini App, website and mobile app, with a matchmaking engine, anonymous message relay and voice/video calls (Agora, LiveKit, ZegoCloud).

- Designed an AI safety layer: profile-photo verification, fake-persona and catfish scoring, and real-time chat moderation, served through a Cloudflare Worker API.
- Built a multi-provider AI manager (OpenAI, Claude, Gemini) with versioned prompts, per-call token, cost and latency logging, and daily usage limits that fail closed.
- Built a Telegram group admin bot with AI moderation, anti-raid protection, appeals, scheduled messages and analytics, using retrying event workers.
- Added a payment hub with a double-entry ledger, country-based feature flags and a locked cron scheduler; ran slow-query and index audits.

AI Support Agent and CRM Platform | Cloudflare Workers, Workers AI, Vectorize, D1, KV, CodeIgniter 4

- Built an AI support agent on Cloudflare Workers running Llama 3.3 70B, answering from a knowledge base through RAG (bge embeddings stored in Vectorize) with conversation history in D1.
- Gave the agent tool calling to act inside the CRM (e.g. create tickets) and streamed step-by-step progress to the chat widget as NDJSON; added a public website chat widget with contact and newsletter tools.
- Automated CRM workflows through signed webhooks: AI triage of new tickets, AI lead qualification and overdue-invoice follow-ups, processed in the background with waitUntil.
- Secured the API with scoped per-user keys, KV caching and rate limiting; built an OAuth MCP server so AI clients (Claude, Codex, Cursor) can use the CRM with each user's permissions.
- Extended the CodeIgniter 4 CRM (clients, projects, estimates, invoices, contracts, tickets, leads) with WhatsApp integration and real-time WebSocket (ReactPHP, ZeroMQ) and Firebase push notifications.

Evonex – Agency Operations Platform | PHP 8.2, MySQL, PHPMailer, OpenAI / Gemini / Claude APIs

- Built and documented (PRD, database spec, architecture) an agency workspace for leads, proposals, projects, invoicing, renewals, a client portal and BI dashboards, with an AI assistant panel.

WhatsApp API Backend | CodeIgniter 4, PHP 8.5, Ratchet, Redis, Docker

- Developed a WhatsApp API backend with webhooks, WebSockets and Redis, deployed with Docker across Dev, UAT and Production, and documented with Postman collections.

PSMMS – Membership Management System | Core PHP 8.5 MVC, MySQL, Bootstrap 5, PHPUnit

- Built a membership system with admin and member portals and a public website on a framework-free PSR-4 MVC, with database migrations and PHPUnit tests.

Attendance Management System | PHP, MySQL, PhpSpreadsheet

- Built a system to manage students, faculty, subjects, attendance, fees and notices, with monthly and subject-wise reports, Excel backups and semester promotion.

Meet – Video Conferencing SaaS | Laravel, Vue.js, Node.js, Socket.io, WebRTC

- Customized a video meeting platform with subscription plans (Stripe, PayPal, Razorpay), two-factor authentication, social login and a ChatGPT assistant.

Spy – SEO Tools SaaS | Laravel 10, Vue.js, MySQL

- Customized an SEO tools platform with keyword tracking, backlink checks, WHOIS lookup and a URL shortener, sold through subscriptions (Stripe, PayPal, Razorpay).

WOG – Crypto Rewards Platform | CodeIgniter 4, MySQL

- Co-developed a faucet and offerwall platform with wallet withdrawals (FaucetPay, CCPayment), referrals, proof-of-work captcha and a request firewall.

SmartStore – E-commerce Store | CodeIgniter, MySQL, Paytm

- Customized an online store with cart, coupons and checkout, and integrated the Paytm payment gateway.

Computer Training Institute Portal | PHP, MySQL, JavaScript

- Built an institute website with admin and student portals for courses, admissions, enrollments, payments, OTP verification and certificates.

Preeti Janawade – Personal Brand Website | HTML, CSS, JavaScript

- Built a client website with booking, newsletter and payment-report forms connected to a backend API, plus WhatsApp, Instagram and Google Maps integration.

ACADEMIC PROJECT

Job Search Portal | PHP, MySQL, jQuery

- Developed a job search web application using PHP and MySQL, with jQuery for the front-end interactions.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Engineering 2022 – 2025
Dream Institute of Technology | CGPA: 6.78

Diploma in Computer Science and Technology 2019 – 2022
Murarai Government Polytechnic | GPA: 8.5